

Autonomous car in Carla

Akademia Górniczo-Hutnicza im. Stanisława Staszica w Krakowie
AGH University of Krakow

Paweł Poręba

Carla simulator



Carla simulator

- A customizable simulator of traffic



Carla simulator

- A customizable simulator of traffic
- Based on unreal engine



Approaches to autonomous driving

Approaches to autonomous driving

- an algorithm

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- an algorithm
- data-based machine learning

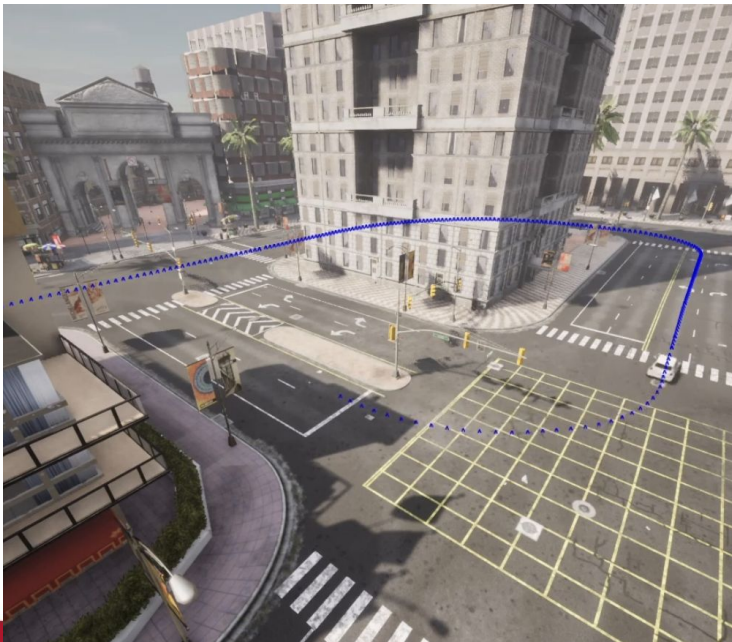
Approaches to autonomous driving

- an algorithm
- data-based machine learning
- reinforcement learning

The data to be processed

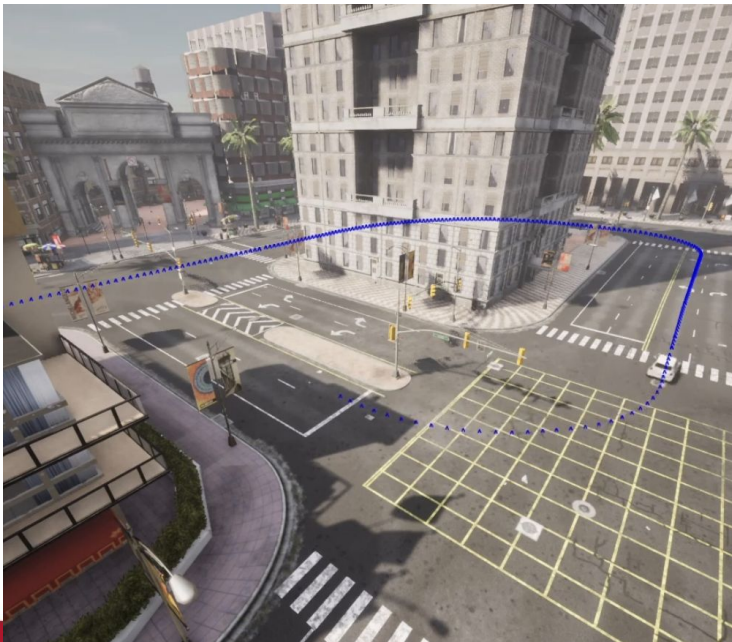
The data to be processed

- The navigation route



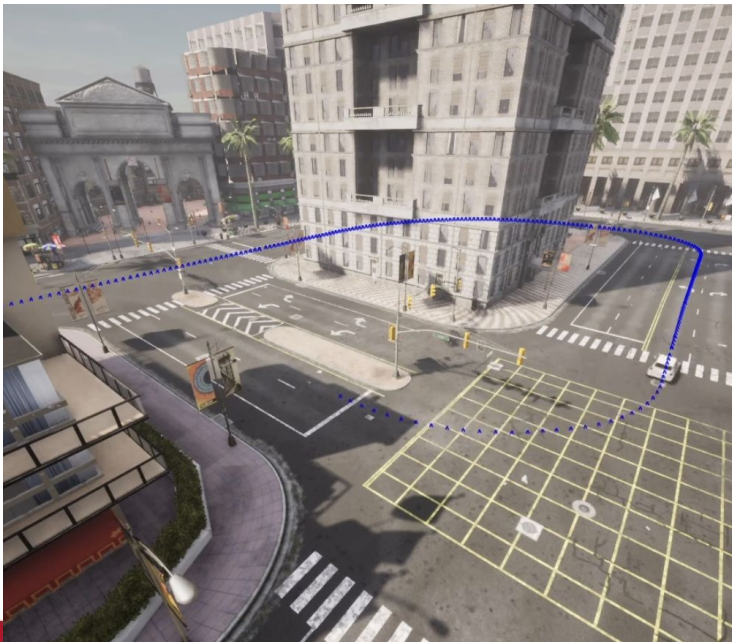
The data to be processed

- The navigation route
- The traffic data



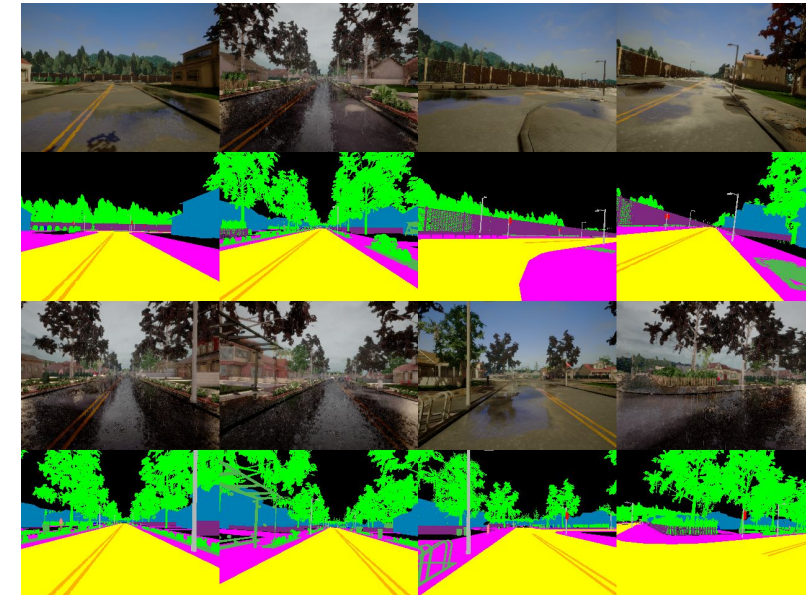
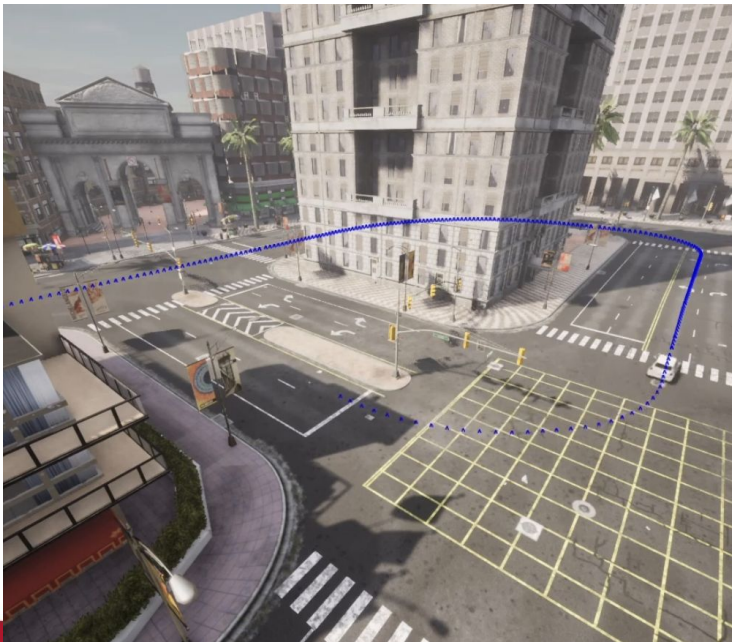
The data to be processed

- The navigation route
- The traffic data
- Car position and state



The data to be processed

- The navigation route
- The traffic data
- Car position and state
- Semantic and non-semantic camera view



Reinforcement learning



Reinforcement learning

- Define rules, rewards and penalties

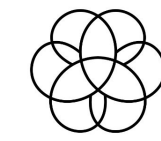
Reinforcement learning

- Define rules, rewards and penalties
- Let the model run and learn

My implementation

Used packages:

- Gymnasium (openai)
- stable_baselines3 (and pytorch)



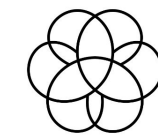
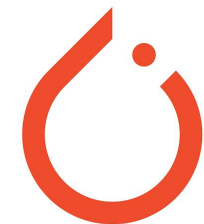
Gymnasium



My implementation

Used packages:

- Gymnasium (openai)
- stable_baselines3 (and pytorch)
- I migrated out of stable_baselines3 to use just pytorch (with the help of a copilot), because stable_baselines3 kept hanging on my PC

 Gymnasium

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- Output data: steering angle, throttle and brake forces

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 - Angle between cars orientation and the next routepoint

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Rewards:

- making distance
- following the route

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Rewards:

- making distance
- following the route

Punishments:

- collisions
- getting farther away from the route

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- First, it generates a lot of car trajectories
- Then, using PPO algorithms, it learns, which is better and in what way

What to do next

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- Expand inputs

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- Handle learning data better
- Expand inputs
- Punish for violating traffic laws

Thanks for the attention